

Kamlesh Kumar

Full-Stack Software Engineer, AI Agents & DevOps

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SUMMARY

Software engineer with 2.5 years of experience. I work full-stack in **TypeScript**, **React**, and **Node.js**, build **AI agents** with tool-calling, and run my own deployments with **kubernetes** and **CI/CD**. Most recently I was the only engineer on a live platform used by 1,600+ users and 250+ companies. Before that, a PCI DSS Level 1 payment gateway and the open-source Apache AGE database.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C/C++, SQL
- **Frontend:** React, Next.js, component architecture, dashboards, responsive UI
- **Backend:** Node.js, REST API design, FastAPI, Spring Boot, microservices, OAuth 2.0, JWT, webhooks
- **AI & LLM:** AI agents, Model Context Protocol (MCP) tool-calling, LangChain, OpenAI API, LLM fine-tuning (LLaMA-2 7B, DistilGPT-2), PyTorch, Hugging Face, prompt engineering
- **DevOps & Cloud:** Docker, Kubernetes (GKE), Terraform, AWS CDK, Infrastructure-as-Code, CI/CD, GitHub Actions, Jenkins, Linux, Git; AWS, Google Cloud
- **Databases:** PostgreSQL, MongoDB, MSSQL, Supabase; schema design, indexing, query optimization
- **Ways of Working:** Agile, code review, automated testing, production support, client-facing delivery

WORK EXPERIENCE

- **Full-Stack Developer (HR-Tech)** Espoo, Finland
XSTRYV, Full-time Mar 2026 – Jun 2026
 - Took ownership of the codebase from day 1, delivering **67 commits in 4 months** across the frontend, server, and data layers, and later onboarded two new developers onto the platform.
 - ◊ **Application architecture:** Built and extended feature modules for jobs, chat, admin, onboarding, and dashboards in **Next.js (App Router)** with React, TypeScript, Tailwind, and Radix UI, backed by a typed service layer.
 - ◊ **Server layer:** Implemented server-side logic through **Next.js Server Actions** covering authentication, messaging, admin approval flows, bulk job operations, and candidate data export to Excel.
 - ◊ **Data layer:** Designed and shipped **14 PostgreSQL migrations** with **Prisma** covering company-scoped chat, review statuses, rejection reasons, and program enrolment.
 - ◊ **Authentication:** Worked across **Supabase Auth** OAuth and email flows, unified separate company and talent sign-in into a single entry point, and hardened one-time verification links against automated email scanners.
 - ◊ **Async processing & caching:** Used **Inngest** for background workflows, **Upstash Redis** for caching, and **Resend** for transactional email, including debounced notifications for unread messages.
 - ◊ **Reliability:** Configured **Sentry** error monitoring and added audit logging through a dedicated service.

Tech stack: Next.js, React, TypeScript, PostgreSQL, Prisma, Supabase Auth, Inngest, Tailwind
- **Full Stack Engineer (Fin-Tech)** Ireland, Remote
Datapulse Technologies, Full-time Jun 2024 - July 2025
 - Full Stack Engineer on a PCI DSS Level 1 payment gateway (25+ years legacy), operating in 20+ currencies and processing hundreds of millions of Euros; collaborated with teams across Ireland, Cyprus, and Pakistan, and modernized core services.
 - ◊ **Google Pay Integration:** Coordinated with Google to onboard the gateway as a participating processor (listed on Google's site); integrated Google Pay with full merchant support (notifications, webhooks, emails)
 - ◊ **Apple Pay Integration:** Contributed to completing Apple Pay integration into the core gateway.
 - ◊ **APIs for Mobile App:** Designed/implemented PCI-compliant REST APIs secured with JWT; enabled direct mobile integration with the core gateway, improved data consistency, and unified UX across platforms.
 - ◊ **11% Throughput Improvement:** Optimized queries/workflows by understanding gateway architecture, codebase, and database design; reduced end-to-end processing time by 11%.
 - ◊ **Live Gateway Issue Resolution:** Worked directly with merchants on white-labels; resolved notifications, webhooks, risk rules, and performance issues and deployed changes to production independently.

Tech stack: Typescript, Java, Spring Boot, RESTful APIs, YugaByte, Docker, JWT Authentication

• **Software Engineer Intern, Open Source**

South Korea, Remote

Bitnine Global Inc, Part-time

Nov 2022 – Nov 2023

- Contributed to **Apache AGE**, an open-source graph database built on PostgreSQL, coordinating asynchronously with maintainers in another time zone.
 - ◊ Built high-availability infrastructure in C/C++ with load balancing and read/write splitting, working close to PostgreSQL internals.
 - ◊ Developed REST APIs in Go for the database tooling layer.
 - ◊ Built a Database-as-a-Service web application in React and Node.js running on cloud VMs.

PROJECTS

- **AI Agent with MCP Tool-Calling:** Developed an agent that acts through voice and text conversation, using the **Model Context Protocol** for tool-calling so it can securely read and act on a user's calendar and context. Delivered through a Telegram Mini App and the web, with low-latency streaming through the OpenAI Realtime API. [GitHub](#)
Tools: TypeScript, Next.js, MongoDB, MCP, OpenAI Realtime API
- **Cloud Deployment Automation (Infrastructure-as-Code):** Automated a full cloud deployment as versioned, reproducible code covering containers, load balancer, CDN, storage, and networking, all provisioned by one command. Wrote cross-platform scripts to deploy, verify, and tear the environment down in minutes. [GitHub](#)
Tools: AWS CDK, Terraform, Docker, ECS Fargate, CI/CD, TypeScript
- **Search Engine over 6 GB of Data:** Implemented a search engine following **Google's** information retrieval paper, returning multi-word queries over **6 GB** of data in **under 1 second** through custom indexing and query optimization. [GitHub](#)
Tools: Python, NLTK
- **Compiler for a Custom Language:** Wrote a compiler that translates a custom language down to native x86-64 assembly, implementing the tokenizer, parser, type checker, intermediate representation, and code generation by hand, packaged with a full test suite. [GitHub](#)
Tools: Python, Docker, x86-64 Assembly

EDUCATION

• **Master of Science in Computer Science**

Sep 2025 – Present

University of Helsinki (GPA: 4.9/5)

Helsinki, Finland

- **Coursework:** Full-Stack Development, DevOps, Distributed Systems, Introduction to Machine Learning, Engineering of Machine Learning.

• **Bachelor of Science in Computer Science**

Sep 2020 – Jun 2024

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

- **Core Courses:** Data Structures & Algorithms, Operating Systems, Databases, Networks, Distributed Computing, OOP.

CERTIFICATIONS

- **DevOps with Kubernetes:** Kubernetes, Google Kubernetes Engine, GitOps, cluster management – **University of Helsinki MOOC**, Ongoing
- **DevOps with Docker:** Docker, Docker Compose, networking, volumes, multi-stage builds – **University of Helsinki MOOC**, 2026
- **Java Spring Framework 6 with Spring Boot 3:** Spring Boot, JPA, Security, Docker, microservices – **Udemy**, 2024
- **PostgreSQL Indexes:** Indexing and query performance – **Percona University**, 2023
- **IELTS, Band 7.0:** Professional working proficiency in English, written and spoken

LANGUAGES

- English (full professional, IELTS 7.0) | Finnish (basic, improving)

LEADERSHIP & VOLUNTEER

- **United Nations Millennium Fellowship, Campus Director (2023):** Led **17** fellows across **5** projects. Appointed Campus Director after selection into the top 1% of applicants. [Profile](#)
- **Slush Volunteer (2025):** Supported Founder Days at Slush, helping coordinate a gathering of 1,500+ founders.